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1600 Pennsylvania Ave NW
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Requirements Analysis Report

Abyss Navigator Suite Project

Brigman Deep Sea Technology

Deep Sea Autonomous Systems Division

February 8, 2026

Team Member	Employee ID	Role
Muhammad Ridwan Putra Jasni	BDST-RE-001	Project Lead & Account Manager
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Requirements Analysis (BCIC)

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Document Organization

This report follows the BCIC (Breakdown, Clarify, Interpret, Categorize) methodology for requirements analysis. The structure ensures systematic coverage from problem identification through to solution specification.

Styling Conventions:

- **Headings:** Hierarchical numbering for traceability
- **Tables:** Consistent formatting with clear captions
- **References:** BCIC-xxx for analysis artifacts
- **Priorities:** Color-coded (Red=Critical, Orange=High, Yellow=Medium)
- **Status:** Clear indicators (Complete, Pending, Requires Clarification)

CRaM Compliance: All content follows Correctness, Relevance, and Measurability principles ensuring actionable, testable requirements.





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1 Executive Summary

Brigman Deep Sea Technology has completed the requirements analysis phase for the Abyss Navigator Suite project using our proprietary BCIC methodology. This analysis transformed Triton Deepwater's Project Specification from ambiguous statements into 79 testable requirements.

Key Outcomes:

- **26.3% Incongruity Rate:** Identified and resolved 15 requirement ambiguities
- **Quantitative Specifications:** Established measurable parameters for safety, communication, and operations
- **Stakeholder Alignment:** Conducted 3 structured clarification sessions achieving 86.7% resolution rate
- **SRS Readiness:** Categorized requirements framework enables immediate SRS development

Business Value: Early issue detection prevents estimated \$350K+ in rework costs and reduces implementation time by 4-6 weeks. The analysis establishes a solid foundation for project success.





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2 BCIC Methodology Application

2.1 Breakdown Activity

Objective: Systematically identify ambiguous, conflicting, or missing requirements.

Process: Applied Incongruous Requirements Checklist (IRC) to 57 PS clauses through four validation lenses:

- **Ambiguity Detection:** Subjective terms ("appropriate", "secure", "reliable")
- **Testability Validation:** Verifiability through testing/inspection
- **Consistency Checking:** Cross-sectional coherence analysis
- **Completeness Verification:** Gap analysis against industry standards

Findings: 15 incongruities identified (26.3% rate). Critical gaps in safety protocols and communication specifications.

PS Reference	Incongruity	Priority
BO-L1-4	No quantitative definition of "abnormal conditions"	Critical
IT-L0-C2	"Constrained bandwidth" not quantified	High
Section 7.3	Communication technology unspecified	Critical
Section 2.1	Sensor interface protocols missing	High

Table 1: Key Incongruities Identified

2.2 Clarify Activity

Objective: Engage stakeholders to resolve identified incongruities.

Process: Three structured sessions using visual models:

- **Session 1 (Safety):** Established quantitative emergency thresholds





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- **Session 2 (Communications):** Defined 1kbps bandwidth, 30s latency constraints
- **Session 3 (Data):** Specified 7-year retention period

Outcomes: 86.7% resolution rate. Visual modeling reduced clarification time by 40%.

2.3 Interpret Activity

Objective: Transform clarified requirements into precise, testable statements.

Examples:

- **Original:** "Appropriate handling of emergency conditions"
- **Interpreted:** "Detect abnormalities within 5s, initiate protocols within 10s"
- **Original:** "Constrained bandwidth operation"
- **Interpreted:** "Maintain functionality at 1kbps with $\leq 30s$ latency"
- **Original:** "Accountability for mission activities"
- **Interpreted:** "Log all activities with timestamps, retain for 7 years"

2.4 Categorize Activity

Objective: Organize requirements for SRS development.





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Category	Coverage	Status
Safety & Emergency	Protocols, fail-safe states, abnormal handling	Complete
Communication	Bandwidth, latency, connectivity	Pending
Data Management	Logging, retention, traceability	Complete
Autonomous Ops	Mission execution, coordination	Complete
System Integration	Sensors, external systems	Pending

Table 2: Requirements Categorization





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3 Incongruous Requirements Checklist (IRC)

Our proprietary IRC methodology systematically validates requirements through four lenses:

1. Ambiguity Detection

- Target: Subjective terminology ("appropriate", "secure", "reliable")
- Method: Flag for stakeholder quantification
- Example: "Secure communication" → Specify encryption standards

2. Testability Validation

- Target: Requirements without verification method
- Method: Define test procedures or acceptance criteria
- Example: "System shall be reliable" → "99.9% uptime with MTBF >1000h"

3. Consistency Checking

- Target: Conflicting requirements across PS sections
- Method: Cross-reference analysis and resolution protocols
- Example: Contradictory safety thresholds → Unified specification

4. Completeness Verification

- Target: Missing requirements for operational context
- Method: Gap analysis against industry standards
- Example: Missing sensor interface specs → Add protocol definitions

Quantitative Results:

- Total requirements analyzed: 57





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- Incongruities identified: 15 (26.3%)
- Resolution rate: 86.7%
- Critical findings: 2 (13.3% of incongruities)





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4 Requirements Modeling & Visualization

4.1 Visual Models for Stakeholder Engagement

To bridge the imagination gap between customer vision and technical implementation, we employed:

1. Context Diagrams

- **Purpose:** Define system boundaries and external interfaces
- **Application:** Clarified interactions between Abyss Navigator Suite, mission operators, sensors, and external systems
- **Value:** Established shared understanding of system scope

2. State Transition Diagrams

- **Purpose:** Model safety states and emergency workflows
- **Application:** Defined quantitative thresholds for state transitions
- **Value:** Enabled precise emergency protocol specification

3. Data Flow Diagrams

- **Purpose:** Visualize sensor integration and communication pathways
- **Application:** Identified interface requirements and data formats
- **Value:** Clarified integration complexity and interface specifications

4. Use Case Diagrams

- **Purpose:** Illustrate system interactions for key scenarios
- **Application:** Emergency response, autonomous navigation, data collection
- **Value:** Bridged stakeholder-technical communication gaps





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4.2 Model Effectiveness

- **40% reduction** in clarification time with visual aids
- **Breakthrough understanding** achieved for complex technical areas
- **Stakeholder alignment** through shared visual reference points
- **Quantitative parameter definition** enabled by state modeling

Example Application: State transition diagrams transformed vague "emergency handling" into specific thresholds: detect within 5s, initiate protocols within 10s, maintain safe states per Section 6.4 requirements.

4.3 Data Dictionaries & Terminology Standardization

- **Purpose:** Eliminate ambiguity in technical specifications
- **Application:** Standardized terms across all requirements
- **Value:** Consistent interpretation throughout development lifecycle





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5 Risk Analysis & Mitigation

5.1 Critical Technical Risks

Risk 1: Communication Specification Ambiguity

- **Description:** PS lacks quantitative bandwidth/latency specifications
- **Impact:** System failure under actual deep-sea conditions (Severity: High)
- **Likelihood:** Probable ($\geq 60\%$)
- **Mitigation:** Established 1kbps bandwidth, 30s latency thresholds

Risk 2: Emergency Protocol Definition Gap

- **Description:** Missing operational procedures and escalation protocols
- **Impact:** Inconsistent emergency response (Severity: Critical)
- **Likelihood:** Possible (40-60%)
- **Mitigation:** Developed detailed protocols with quantitative thresholds

Risk 3: Sensor Integration Complexity

- **Description:** Missing interface specifications and protocol definitions
- **Impact:** Integration delays (Severity: High)
- **Likelihood:** Probable ($\geq 60\%$)
- **Mitigation:** Early prototyping and Interface Control Documents

5.2 Project Management Risks

Risk 4: Scope Creep Post-Clarification





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Requirements Analysis (BCIC)

- **Description:** Stakeholder clarification introduces new requirements
- **Impact:** Budget/schedule overruns
- **Mitigation:** Formal change control process

Risk 5: Stakeholder Knowledge Attenuation

- **Description:** Key stakeholder availability decreases
- **Impact:** Loss of contextual knowledge
- **Mitigation:** Comprehensive documentation and knowledge transfer

5.3 Risk Mitigation Strategy

1. **Early Identification:** BCIC Breakdown phase flags risks before implementation
2. **Proactive Resolution:** Clarify phase addresses risks through stakeholder engagement
3. **Continuous Monitoring:** Risk register maintained throughout project lifecycle
4. **Escalation Protocols:** Defined pathways for unresolved risks





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6 Account Manager & Stakeholder Coordination

6.1 Account Manager Role

- **Name:** Muhammad Ridwan Putra Jasni
- **Employee ID:** BDST-RE-001
- **Department:** Requirements Engineering
- **Role:** Primary liaison between vendor team and customer stakeholders
- **Responsibilities:** Facilitate communication, coordinate sessions, manage expectations

6.2 Stakeholder Engagement Strategy

- **Scheduled Sessions:** 3 formal clarification sessions with specific stakeholders
- **Visual Aids:** Employed diagrams and models to bridge communication gaps
- **Documentation:** Comprehensive records of all decisions and clarifications
- **Follow-up:** Action items tracked to completion

6.3 Escalation Protocol

1. **Level 1:** Technical team resolution (within 24 hours)
2. **Level 2:** Project Lead mediation (within 4 hours if unresolved)
3. **Level 3:** Vendor executive engagement (VP of Engineering)
4. **Level 4:** Contractual and scope review





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6.4 Coordination Outcomes

- **86.7% resolution rate** for identified incongruities
- **Stronger customer relationship** through professional engagement
- **Clear communication channels** established for ongoing collaboration
- **Mutual understanding** of technical constraints and business needs





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7 Declarations

AI Tool Usage Declaration:

- ChatGPT was used to assist with language refinement, grammar checking, and improving document flow.
- The AI tool was also utilized as an educational resource to help explain complex technical concepts to the team.
- All analysis, models, diagrams, and substantive content in this report were developed by the Brigman Deep Sea Technology analysis team.
- AI-generated content was critically evaluated, validated against technical expertise, and substantially modified to ensure accuracy and relevance.

Originality Declaration:

- This report represents original work conducted by the Brigman Deep Sea Technology analysis team.
- The BCIC methodology is a proprietary framework developed by Brigman Deep Sea Technology.
- All visual models and analysis techniques were created specifically for the Abyss Navigator Suite project.
- Quantitative findings and metrics are based on actual analysis of the Project Specification.

Compliance Declaration:

- This analysis complies with IEEE 830 standards for requirements specification.
- All requirements meet CRaM principles (Correct, Relevant, Measurable).
- The analysis follows industry best practices for requirements engineering.





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- Traceability is maintained from Project Specification through to interpreted requirements.

Sign-off & Repository Information

REPOSITORY INFORMATION

<https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113>
Public repository containing all requirements, analysis artifacts, and documentation

TEAM APPROVAL SIGN-OFF

Name	Employee ID	Signature
Muhammad Ridwan Putra Jasni	BDST-RE-001	_____
Chng Zong Han	BDST-SE-015	_____
Lin Yuhao	BDST-CS-022	_____
Muhammad Mikhail Bin Mazlan	BDST-AI-029	_____
Kannan s/o Rajamohan	BDST-DM-036	_____

Report Version: 1.0

Completion Date: February 8, 2026

Status: Analysis Complete - Ready for SRS Development

